Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces

Class test 1 **19 March 2024**

Time : 18:00 – 20:00 **Total marks: 40**

1. Let d_E be the Euclidean metric on $\mathbb{R}$ and d_D the discrete metric on $\mathbb{R}$.
   Let $A = \{-1\} \cup (0,4) \cup (4,7]$.

   (a) In $(\mathbb{R}, d_E)$ write down $\text{int}(A)$. (No proof required.)

   (b) In $(\mathbb{R}, d_E)$ write down $\overline{A}$. (No proof required.)

   (c) In $(\mathbb{R}, d_E)$ write down $\text{bd}(A)$. (No proof required.)

   (d) In $(\mathbb{R}, d_D)$ write down $\text{int}(A)$. (No proof required.) [4]

2. Suppose (X, d) is a metric space. Define ρ by:

$$\rho(x, y) = \min\{1, d(x, y)\} \text{ for all } x, y \in X$$

   (a) Prove carefully that ρ is a metric on X.

   (b) Now let $(X, d) = (\mathbb{R}, d_E)$. In $(\mathbb{R}, \rho)$, find the open ball $B(0; \frac{1}{2024})$. (No proof required.)

   (c) Still let $(X, d) = (\mathbb{R}, d_E)$. In $(\mathbb{R}, \rho)$, find the open ball $B(0; 2024)$. (No proof required.) [8]

3. Prove or disprove:

   (a) If A and B are subsets of a metric space (X, d), then $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

   (b) If A and B are subsets of a metric space (X, d), then $\text{int}(A \cup B) = \text{int}(A) \cup \text{int}(B)$.

   (c) If A is a subset of a metric space (X, d), then $\text{int}(\overline{A}) = \text{int}(A)$.

   (d) If A is a subset of a metric space (X, d), then $\text{bd}(\text{bd}(A)) = \text{bd}(A)$.

[12]

4. Read the following theorem and proof, and then answer the questions that follow.

Theorem Let (X, d) be a metric space and $A \subseteq X$. If A is open, then, for all $a \in A$, there exists $r > 0$ such that $B(a, r) \subseteq A$.

Proof Suppose that A is open, i.e. that A' is closed. (1)
We use proof by contradiction. (2)
So we assume there exists $a \in A$ such that, for all $r > 0$, $B(a, r) \nsubseteq A$. (3)
So, for each $n \in \mathbb{N}$, there exists $x_n \in B(a, \frac{1}{n})$, but $x_n \notin A$. (4)
Then $x_n \to a$. (5)
However, $x_n \in A'$ for each n. (6)
So $a \in A'$. (7)
This gives a contradiction. (8)

(a) Explain clearly how line (4) follows from the previous working.

(b) Show that $x_n \to a$, as claimed in line (5).

(c) How do we know that $x_n \in A'$, as claimed in line (6)?

(d) How do we know that $a \in A'$, as claimed in line (7)?

(e) What is the contradiction referred to in line (8)?

[12]

5. (a) Prove or disprove: If d is a metric on a set X, then d^2 is also a metric on X. State clearly whether the result is true or false.
(Note that $d^2(x, y) = (d(x, y))^2$).

(b) Let F and K be closed subsets of a metric space (X, d). We define:

$F \ll K$ iff there exists a closed subset H of X such that $F \cap H = \emptyset$ and $H \cup K = X$.

Prove that, for F a closed set in a metric space X,

$$F = \bigcap \{K \in P(X) \mid K \text{ is a closed subset of } X \text{ and } F \ll K\}.$$

[4]

THE END